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Enhancing Design Historical Education Through AI Virtual Characters Role-Playing Narratives in Serious Games

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ABSTRACT

The potential of serious games to improve students' engagement and knowledge retention in design historical education has gained significant attention in recent years. This study proposes a learning strategy integrating AI virtual characters into a role-playing serious game designed to teach design history. Utilizing the Arts and Crafts Movement as the narrative foundation, the game employs AI-driven characters to provide adaptive guidance, interactive storytelling, and personalized feedback. A quasi-experimental design was adopted with two groups of college students (N = 60): an experimental group using the AI-supported game and a control group playing the same game without AI integration. Results show that the inclusion of AI virtual characters significantly enhances students' engagement and knowledge acquisition compared to the traditional game. These findings highlight the potential of AI-powered serious games to transform historical education by fostering active participation and deep learning.

KEYWORDS

Design Education, Design History, Arts and Crafts Movement, Patterns, AI, AI Virtual Characters, Serious Games, Education

INTRODUCTION

In recent years, the application of artificial intelligence (AI) technologies in education has expanded significantly (Abulibdeh et al., 2024). These advancements have demonstrated notable potential in areas such as virtual reality (VR), natural language processing, and personalized learning systems. As educational approaches increasingly shift from traditional didactic methods to models that emphasize active learning and student engagement, AI-driven technologies offer new possibilities for educational innovation (Mao et al., 2024). Serious games, which integrate education and entertainment features, have gained widespread attention due to their ability to enhance learning outcomes through interactivity and immersive experiences (Arif et al., 2024). This approach is particularly valuable

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in knowledge-intensive fields like history education, where it holds significant potential for both research and practical applications (Kara, 2024).

Design history education has long faced challenges, including low student engagement, uninteresting content, and difficulties in internalizing knowledge (Fareed et al., 2024). Traditional methods often rely on one-way knowledge delivery, which struggles to stimulate students' interest in historical content or foster a deeper understanding (Brandl & Schrader, 2024). By creating virtual scenarios that incorporate historical events, design movements, and cultural contexts into interactive gameplay, serious games have not only increased students' interest in learning but also provided a new pathway for experiential knowledge transfer (Agusdinata et al., 2024). However, optimizing the teaching effectiveness of serious games remains an area worth further exploration (Picchiottino et al., 2024). Recently, the integration of virtual characters into serious games has offered a novel solution to this challenge (Tolks et al., 2024).

Virtual characters, utilizing natural language processing and data-driven personalized feedback, enable real-time interaction with players. They provide learners with deeper guidance and support (Campitiello et al., 2024). Studies suggest that virtual characters significantly enhance learning motivation, improve the learning experience, and offer adaptive instructional recommendations (Papagiannakis et al., 2024). The role-playing narrative mechanism further reconstructs the interaction between learners and historical contexts, presenting complex design history knowledge in a more engaging and interactive format (Capecchi et al., 2024).

This study developed a serious game centered on the narrative of the arts and crafts movement, integrating virtual characters to examine their role in design history education. The research was conducted within a university design course, focusing specifically on the arts and crafts movement unit to evaluate students' engagement and knowledge acquisition. In the game-based learning process, students assumed roles that allowed them to interact with virtual characters, collect key historical and design-related information, and connect these insights with broader cultural and historical contexts. This approach helped students organize and internalize knowledge gained through gameplay, complementing the content learned in their course materials. The research focused on two key questions:

Q1: Does integrating AI virtual characters improve students' knowledge compared to traditional serious games?
Q2: Does integrating AI virtual characters enhance student engagement compared to traditional serious games?

Through this investigation, the study aimed to provide fresh perspectives on advancing digital and intelligent approaches to design history education. Additionally, it explored the broader potential of using educational technology to enhance serious games and improve learning outcomes.

LITERATURE REVIEW

Current Status of AI Virtual Characters in Game Learning

The use of AI virtual characters in-game learning has been a growing area of research in recent years. For example, Fitria (2021) explored the implementation of the "Gather Town" game platform as an alternative tool for English language learning during the COVID-19 pandemic. This study highlighted how students perceived Gather Town as a platform with graphics resembling the Harvest Moon game, where they could assume a character and interact with peers in a virtual environment. Building on the theme of character simulation, Rodrigues and Nóbrega (2023) investigated the use of imitation learning alongside game engine physics to develop AI-driven character interactions in

virtual environments. Their research utilized game engines like Unity3D to create three-dimensional sensing characters capable of interacting with AI peers and the environment.

Similarly, Kim and Baylor (2006) introduced the application of pedagogical agents (PALs) as learning companions in a computer learning environment and emphasized that PALs can enhance learning in a computer environment by simulating social interaction, improving learners' interest and self-efficacy, and their affective expression and gender characteristics also affect learners' motivation and expectations. The study underscored the importance of agent design in intelligent tutoring systems and multi-agent learning environments. Expanding into sports-related gaming, Bai et al. (2021) proposed a framework for enhancing emotional experiences in VR volleyball games. They constructed emotionally responsive virtual characters using data sets derived from real-world volleyball game videos. Their data-driven framework demonstrated the potential to significantly improve emotional engagement and participation in VR games. Furthering the exploration of perceptive AI characters, Rodrigues and Nóbrega (2023) examined methods for three-dimensional perceptual simulation using game engines. The study evaluated the integration of external simulation libraries, reinforcement learning, and imitation learning algorithms to train perceptive characters across diverse scenarios, comparing the effectiveness of different approaches.

Together, these studies underscored the increasing integration of AI and machine learning in game-based education. They highlighted the potential of AI virtual characters to personalize learning experiences and improve engagement. However, while these advancements are promising, Tiwari (2023) emphasized the need for further research and ethical considerations to better understand the opportunities and challenges AI presents in educational contexts.

Narrative and Interaction Mechanics in Serious Game Design

Serious games are increasingly being used for diverse purposes, including education, mental health support, and cognitive development. Carlier et. al. (2020) focused on empowering children with autism spectrum disorder and their parents through a serious game designed to reduce stress and anxiety. This highlights the potential of serious games in addressing specific challenges faced by individuals with autism. Viudes-Carbonell et. al. (2021) emphasized the importance of iterative design in serious games to ensure engagement and effectiveness, as many attempts may result in non-engaging games that fail to meet their intended purpose.

In the context of education, Govender and Arnedo-Moreno (2021) conducted a review of game design elements used in digital game-based language learning. The study highlighted the popularity of gamification as an approach and aimed to categorize observations about game elements used in published papers. This underscores the significance of incorporating engaging game mechanics in serious games to enhance learning outcomes. Similarly, Ibrahim et. al. (2021) explored gamification design elements for mental health support, emphasizing the need to identify significant gamification elements that can influence mental health states in game design.

Building on these insights, Zhang et al. (2021) proposed a serious game design for cultural heritage education based on the experiential learning cycle model. The game aimed to strengthen players' construction of cultural heritage knowledge through narrative, interaction, and knowledge cards. This demonstrates the potential of narrative and interaction mechanics in serious game design to facilitate active learning and engagement. Pescarin and Martinez Pandiani (2022) analyzed the impact of story structure, meaningfulness, and concentration in serious games, highlighting the importance of these factors in enhancing knowledge acquisition and engagement in educational games. Their findings emphasize the need for well-crafted narratives and meaningful interactions.

The role of narrative structure and interaction mechanics is further emphasized by Pescarin and Martinez Pandiani (2022), who found that meaningful storytelling and concentrated tasks enhance engagement and knowledge retention. Moreover, the incorporation of AI characters aligns with findings by Zhu et al. (2021), who explored how AI-driven characters can interact effectively within game environments.

Needs and Challenges of Arts and Crafts History Education

Design history education encompasses the study of design movements, influential designers, technological advancements, cultural and social influences, and the evolution of aesthetics across different time periods. It examines how design has shaped and been shaped by historical contexts, including industrial, graphic, fashion, interior, and product design. This field includes critical analysis of styles, materials, production methods, and the role of design in society, helping students understand design's impact and legacy. The arts and crafts movement in design history education has been a topic of interest in various studies related to art education. Lovano-Kerr (1985) discussed the implications of discipline-based art education for teacher education programs, highlighting the need for changes in pre-service classroom and art teacher education. Fehr (1993) explored the power dynamics in education, art, and culture, shedding light on the sources that shaped his perspective. Miele (2005) delved into the origins of the architectural conservation movement in the United Kingdom, touching upon the relation of the arts and crafts movement to Victorian ideas of heritage. Chalmers and Dancer (2008) examined early influences on art education for boys, focusing on the work of Ernest Thompson Seton. Digranes (2009) traced the development of the Norwegian school subject art and crafts, emphasizing the shift toward democratic design and citizenship. Quílez (2018) analyzed the incorporation of female students in fine arts and design education in Spain. Guney (2018) discussed the effects of economic and technological changes on design professionals and education. These studies provide insights into the historical, cultural, and pedagogical aspects of the arts and crafts movement in design history education.

Design history education worldwide has evolved significantly, with diverse approaches tailored to different cultural and academic contexts. For example, Kokko (2022) identified five major orientations of handicraft learning in higher education across the United States and Europe. These include educational handicraft, traditional handicraft, critical handicraft, cultural heritage, and design-based handicraft. The study emphasizes the unique value of craftsmanship in higher education and analyzes the teaching objectives, student career prospects, and pedagogical methods within each orientation. Similarly, Her (2020) explored the development of contemporary handicraft education in South Korea, tracing its transformation from vague beginnings to specialized programs focused on nurturing elite artists and creators. These studies highlight the evolving role of craftsmanship and its enduring significance in educational contexts.

In China, design history education has also seen notable developments. The T'ou-Se-We Orphan Arts and Crafts Institute in Shanghai, for instance, played a vital role in cultivating visual literacy and enhancing sensory experiences during the Republic of China era. Through innovative tools like magic lantern projectors and films, the institute aimed to preserve and promote traditional crafts (An, 2020). More recently, the use of flipped classroom methods in Chinese arts and crafts history education has garnered attention. Yuan (2019) investigated student attitudes toward this innovative teaching approach, emphasizing its potential to modernize traditional education. As technology advances, the integration of tools like big data has also created opportunities for preserving traditional crafts while adapting to contemporary demands (Song, 2021).

The pandemic further challenged craft pedagogy globally, shifting many programs to remote education. Kouhia et al. (2021) analyzed the effects of this transition, finding that remote teaching tended to be more teacher-centered and task-oriented compared to classroom interaction. These findings underscore the need for innovative strategies to ensure engagement and maintain the quality of design history education.

Against this backdrop, the arts and crafts movement stands out as an essential subject in design history, given its profound influence on cultural and artistic development. Originating in the United States, the movement emphasized craftsmanship and the value of handmade goods. It later spread to the United Kingdom and Russia, leaving an indelible mark on society and culture (Clark, 1972; Kaplan & Boris, 1987; Miele, 2005). Its activities aimed to revive traditional crafts and promote an authentic and meaningful way of life, making it a cornerstone of design history education.

The historical significance of the arts and crafts movement also aligns with broader educational practices. For instance, Cartwright (2024) reflected on the Junior Art Department in Sheffield, England, during the mid-20th century. The Junior Art Department's innovative teaching methods and focus on artistic potential shaped the careers of students and educators alike, providing valuable lessons for contemporary arts and crafts education.

DEVELOPMENT

Game Concept and Narrative

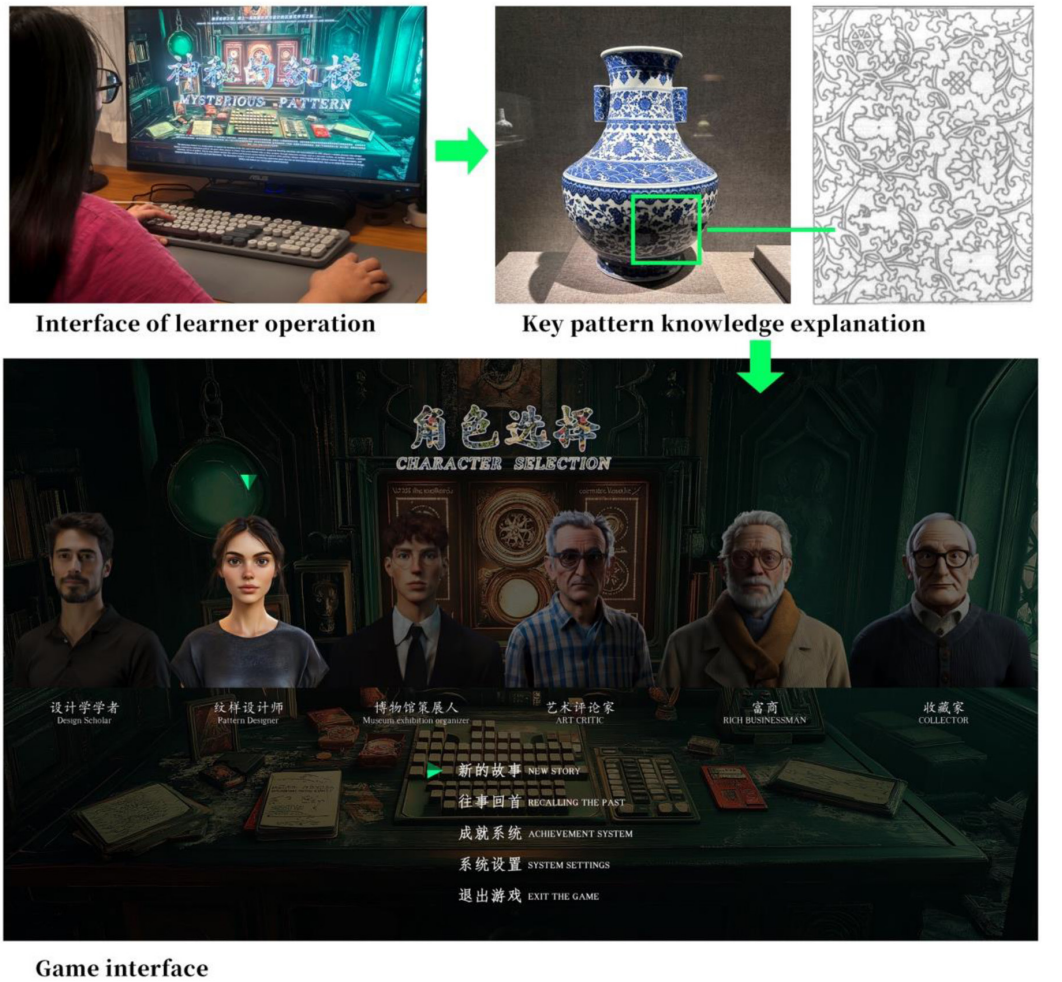
The learning serious game titled "Mystery Patterns" is set against the backdrop of the arts and crafts movement. Through immersive storytelling and role-playing mechanics, it provides a unique design history learning experience. The game focuses on a mysterious pattern, allowing players to choose one of six distinct roles to explore the movement's cultural background, design principles, and historical significance. AI virtual characters guide the experience with feedback, personalized learning paths, and narrative support. Players investigate archives, bid on artifacts, reproduce patterns in workshops, and uncover the cultural and historical stories behind the patterns in an art gallery.

Developed using the Unity engine, the game integrates a knowledge base, non-linear narrative, and real-time feedback to create an interactive learning platform. The game's objectives focus on three aspects. First, it uses storytelling and virtual scenarios to present the social context and design concepts of the arts and crafts movement. Second, AI virtual characters provide role-based tasks and adaptive guidance to align learning with players' interests and abilities. Third, the game encourages active exploration by offering feedback and adjusting task difficulty based on player performance. This approach aims to enhance both learning outcomes and knowledge retention.

Game Structure and Design Details

The game homepage emphasizes the cultural depth of the arts and crafts movement and the narrative atmosphere of mysterious exploration (Figure 1). A deep green color palette establishes a calm and mysterious visual tone, while golden accents and vintage decorative elements highlight the classic characteristics of the movement. Books, sketches, and retro tools displayed on a desk suggest the game's core mechanics—knowledge exploration and puzzle solving. The intricate patterns on the books align with the main storyline, reinforcing the theme of design and historical learning. The background features a classical interior scene with lighting effects, creating a mysterious narrative atmosphere. Accompanied by the tagline, "Uncover the mystery of patterns and embark on an immersive journey through history and design," the cover sparks curiosity and anticipation while establishing the game's cultural context and educational value.

Figure 1. The game "mystery patterns"



In the role selection system, players can choose from six distinct characters (Table 1), each with unique skill sets and personalized AI support. Each role has dynamically generated AI guidance that provides tailored task instructions and real-time feedback based on the chosen character's identity. The role system offers diverse learning pathways to cater to different learning needs.

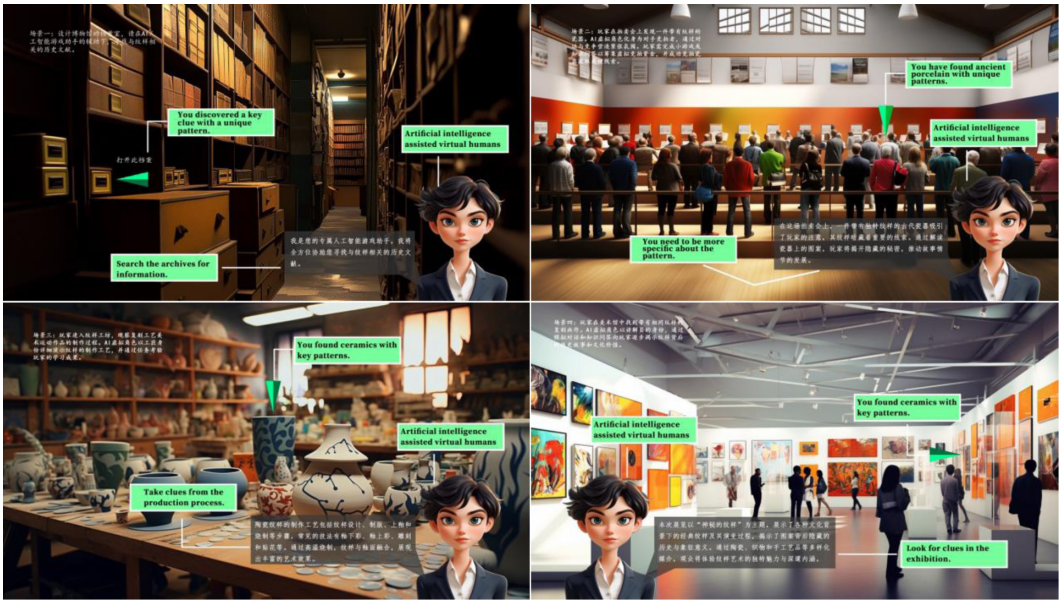
Table 1. Character introduction

No.	Character Name	Character Introduction
1	Scholar	Scholar specializes in analyzing key events and theories in design history. Scholars interpret the principles of the arts and crafts movement and provide valuable insights for exploration tasks. With AI guidance, scholars analyze historical documents and artistic contexts, helping the team build a solid theoretical foundation. Ideal for players seeking an academic understanding of history.
2	Designer	Designer focuses on pattern design and craftsmanship. Designers excel at solving pattern-related puzzles and, with AI support, learn about traditional production techniques. Designers can identify hidden clues in design details. Suitable for players who enjoy hands-on learning and craftsmanship.
3	Museum Curator	Museum curator has extensive knowledge of exhibits and their historical significance. Curators link artifacts to objectives and optimize task routes. With AI support, they manage archives and records efficiently, helping the team focus on essential goals. Best for players interested in historical artifacts and efficient task-solving.
4	Art Critic	Art critic explores cultural meanings hidden in narrative details. With AI assistance, critics uncover the deeper significance of tasks and interpret the symbolic value of artworks. Critics offer unique perspectives and connect clues in innovative ways. Ideal for players interested in stories and cultural contexts.
5	Wealthy Businessman	Wealthy businessman excels in resource management and market strategies. Merchants acquire key items and funds through auctions and trades. With AI assistance, businessman navigate the art market and use economic strategies to achieve goals. Suitable for players who enjoy strategy and resource planning.
6	Collector	Collector focuses on appraising artworks for authenticity and historical value. With AI support, collectors uncover hidden details and task clues while offering cultural insights. Ideal for players interested in detailed analysis and art history.

Note. AI = artificial intelligence.

The game features four core scenarios centered on immersive narratives and interactions with virtual characters (Figure 2). These scenarios use contextualized tasks to facilitate learning and guide players in understanding the design principles, historical background, and cultural values of the arts and crafts movement.

Figure 2. Game scene with the artificial intelligence character



Immersive narratives, combined with realistic stories and historical settings, engage players in specific times and spaces, making the learning process more intuitive. Virtual characters act as mentors or collaborators, offering task feedback and personalized guidance to support players in achieving their learning objectives. As shown in Table 2, each scenario is designed with clear educational goals. Main tasks drive the narrative forward, while side tasks expand knowledge and help players consolidate their learning outcomes.

Table 2. Game scene introduction

Scenario	Context	Task Design
Archive Room of a Design Museum	Players search for historical documents related to patterns with the help of an AI virtual character (archive manager). The AI poses design history questions, such as "What is William Morris's design philosophy?" to activate players' historical knowledge and gradually unlock clues about the patterns' origins.	Answer questions related to the arts and crafts movement to unlock document clues. Use real-time hints provided by the AI for additional support to enhance task completion rates.
Antique Auction	Players discover a patterned porcelain piece at an auction. The AI virtual character, acting as a rival bidder, creates tension through dialogue and competition. Players complete mini-games or knowledge questions to earn virtual bidding funds and successfully acquire the porcelain for key clues.	Complete design-related questions or mini-games to earn bidding funds. Interact with the AI rival to learn about the porcelain's patterns and their historical and cultural background.

continued on following page

Table 2. Continued

Scenario	Context	Task Design
Pattern Workshop	Players enter a workshop to observe the reproduction of Arts and Crafts designs. The AI virtual character, as a craftsman, demonstrates the production process and evaluates players' learning through tasks.	Solve puzzles to access the workshop. Observe the production process and answer questions about the techniques used in pattern-making. The AI provides immediate feedback based on performance and guides players in exploring the historical significance of pattern designs.
Art Gallery	Players find a reproduction painting featuring the same pattern in an art gallery. The AI virtual character, acting as a guide, uses dialogue and questions to reveal the historical stories and cultural values behind the patterns.	Assemble the complete story of the pattern's origins. Complete related questions under the guidance of the AI character.

Note. AI = artificial intelligence.

Technical and Development

The game is developed using the Unity engine and incorporates Generative Pre-Trained Transformer–3.5 to enable the core functions of AI virtual characters, supporting dynamic interaction, contextual generation, and personalized feedback. The development process focuses on educational goals and narrative design, integrating multi-disciplinary knowledge and technologies to ensure a seamless combination of technical and educational content.

The knowledge module integrates historical, cultural, and design-related content from the arts and crafts movement, creating a dynamic knowledge database to support in-game question and answer and learning feedback. Knowledge points are categorized by themes and difficulty levels, allowing the system to dynamically deliver relevant information based on player tasks. For instance, in the museum archive room, the system provides content related to the history of patterns. Using natural language processing technology, players can engage in real-time dialogue with AI characters, enabling deeper knowledge interpretation and expanding learning opportunities while enhancing interactivity and efficiency.

The narrative module generates contextualized content and adjusts story progression dynamically based on player choices and task performance, offering a personalized learning experience. The non-linear narrative engine allows players to explore different learning paths based on their roles. For example, a "design scholar" focuses on design philosophy, while a "pattern craftsman" emphasizes the production process. Task objectives are seamlessly integrated with story progression to enhance immersion. By dynamically generating dialogues with historical figures, players can interact with virtual characters and gain a deeper understanding of the cultural context of the arts and crafts movement.

The art and sound design adhere strictly to the style of the arts and crafts movement. Visual elements incorporate classic patterns, geometric compositions, and natural tones to create an authentic historical atmosphere. Sound design draws inspiration from late 19th-century European music, using classical instruments and environmental sounds to create immersive scenarios. For example, the sound of flipping pages in the archive room or tools in the workshop enhances the authenticity of the experience.

The game development process includes two phases: functional prototype validation and optimization. The Unity engine was used to verify the feasibility of integrating the knowledge database and interaction modules. Generative Pre-Trained Transformer–3.5 was then incorporated to support real-time dialogue and dynamic narrative adjustments. User testing played a critical role throughout development, with the team refining task design, interaction logic, and interface performance based on feedback to balance educational content with engaging gameplay. The development team,

composed of experts in design, history, educational technology, and game development, ensured professional content and technical innovation through interdisciplinary collaboration.

METHOD

For the section of the course focusing on the arts and crafts movement, a quasi-experiment was conducted to evaluate the effectiveness of a serious game-based learning strategy. This system was specifically designed to enhance students' understanding of pattern designs from the movement through role-playing narratives. The study aimed to investigate the impact of this approach on students' knowledge achievement and learning engagement.

In this experiment, students in the experimental group utilized the serious game-based learning system with AI virtual character support, AI–serious game-based learning (AI-SGL), while those in the control group experienced the same games but without AI integration, non- AI–serious game-based learning (NAI-SGL). The course section has been an integral part of the curriculum in the selected universities for several years, ensuring its alignment with existing educational standards.

Research Participants

The participants of this study comprised two classes of college students from two universities located in Jiangsu Province, China. Each class comprised 30 students, reflecting the typical class size in these universities. This setup ensured the inclusion of a diverse representation of students while maintaining alignment with the educational structure prevalent in the region.

For the quasi-experiment, students in one class from one university were assigned to the experimental group, while the other class from another university served as the control group. The experimental group engaged in the AI-SGL system, while the control group used the NAI-SGL. The allocation of groups was designed to minimize biases and provide a balanced comparison.

To ensure consistency, the same instructional objectives, course materials, and serious game scenarios were employed across both groups, differing only in integrating AI virtual characters in the experimental group.

Research Tools

This study employed validated instruments to measure knowledge achievement and learning engagement in "Mystery Patterns" a serious game that integrates AI virtual characters to explore the arts and crafts movement.

Knowledge achievement was assessed through pre-tests and post-tests comprising 20 true-or-false questions. These questions were developed based on the curriculum guidelines of the "History of Modern World Design" course, ensuring alignment with the core learning objectives of the arts and crafts movement. The question set was designed to assess knowledge acquired through the game's immersive learning pathways, including archival research in the museum's document room, pattern analysis during auction events, hands-on experience in the pattern workshop, and cultural discoveries in the art gallery. The questions covered key aspects of the arts and crafts movement that players encountered through their chosen character's perspective, including the movement's cultural context, design principles, historical significance, and the symbolism behind pattern motifs.

The pre-test and post-test contained identical items presented randomly to minimize memory recall bias. This method directly compared students' baseline knowledge and their performance following the game-based learning. To ensure validity, the questions were reviewed by three domain experts in design history and educational game-based learning. Following their feedback, minor modifications were made to refine the wording and ensure alignment with the game's learning objectives. This validation process enhances the reliability of the assessment, making it a robust measure of students' knowledge acquisition.

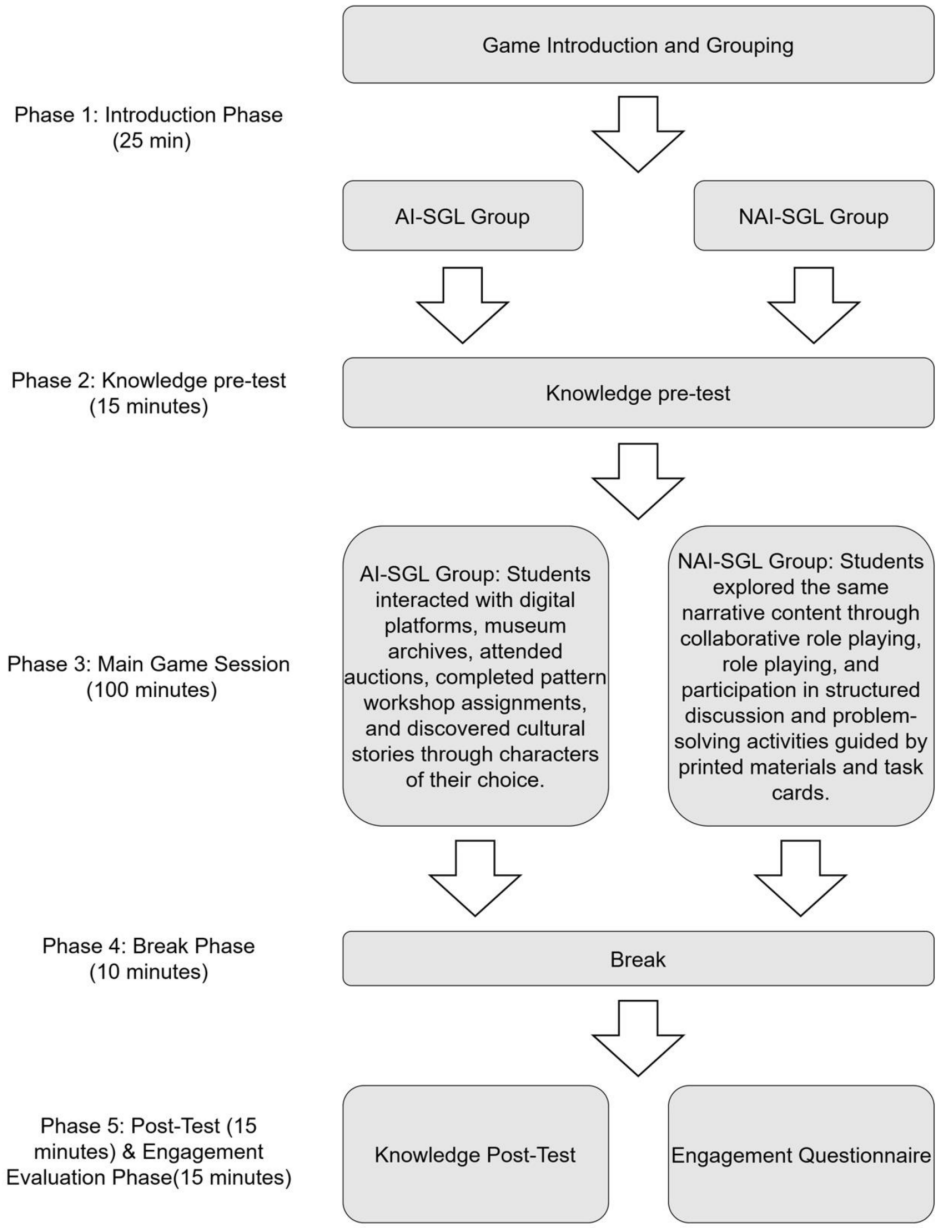
Student engagement was measured using an engagement questionnaire (Reeve, 2013), a 21-item instrument that examines four dimensions: behavioral, agentic, cognitive, and emotional engagement. The engagement questionnaire employs a 7-point Likert scale and has demonstrated robust internal consistency with a Cronbach's alpha of 0.81. Sample items include "When I'm in this game, I try to learn as much as I can." This instrument proved particularly appropriate for the current study, as it captures multiple aspects of engagement relevant to the immersive and narrative-driven nature of role-playing serious games.

The engagement questionnaire was administered as a post-test only, as engagement is a process-dependent variable that requires full exposure to the learning intervention for meaningful assessment. This approach aligns with the theoretical conceptualization of engagement as a process measure rather than a trait measure (Reeve, 2012), allowing students to reflect on their overall involvement throughout the complete learning experience and emphasizing the role of contextual and situational factors in shaping student involvement.

Experimental Procedure

Figure 3 depicts the experimental procedure. The experiment was conducted as a three-hour workshop in a university computer laboratory. All students provided written informed consent before the session commenced.

Figure 3. Research procedure



Note. AI-SGL = artificial intelligence–serious game-based learning; NAI-SGL = non-artificial intelligence–serious game-based learning.

The workshop opened with a 25-minute introductory phase. Students first received a brief study overview (10 min) and the knowledge pre-test (15 min).

Following this, students participated in a 15-min orientation session tailored to their respective groups. Students in the experimental group, AI-SGL, received guided practice with the interactive game interface, focusing on character selection and key gameplay features. Meanwhile, students in the

control group, NAI-SGL, were introduced to the storyline and the procedural aspects of participating in a role-playing narrative, including character assignments and task outlines.

The main gameplay session lasted 100 min. Students in the AI-SGL group interacted with the digital platform to engage with museum archives, attend auction events, complete pattern workshop tasks, and uncover cultural stories through their chosen characters. In contrast, students in the NAI-SGL group explored the same narrative content through collaborative role-playing, assuming character roles, and participating in structured discussions and problem-solving activities guided by printed materials and task cards.

After gameplay, students took a 10-min break before proceeding to the final assessment phase. This 30-min period consisted of the knowledge post-test (15 min) and engagement questionnaire (15 min).

All computer workstations were configured with identical hardware and software to ensure the experiment's validity. A technical assistant was present throughout the session to address any operational issues. Both groups were required to work independently in a quiet and controlled learning environment.

ANALYSIS AND DISCUSSION

Analysis of Knowledge

The purpose of this study was to examine the effects of serious game-based learning on knowledge acquisition and learning engagement. An independent samples t-test was conducted to determine whether there was a significant difference in prior knowledge between the AI-SGL and NAI-SGL groups.

Both groups (AI-SGL and NAI-SGL) had an equal sample size (number [N] = 30), ensuring comparability in the pre-test analysis. As shown in Table 3, the pre-test mean score for the AI-SGL group is 64.50 (standard deviation [SD] = 6.067), while the mean score for the NAI-SGL group is 66.17 (SD = 4.292). The t-test results show no significant difference between the two groups in their pre-test scores, $t(58) = -1.228$, $p = 0.224$ ($p > 0.05$). These findings suggest that the two groups had similar levels of prior knowledge before the learning activity.

Table 3. T-test results of the pre-test

		N	Mean	Standard Deviation	t
Pre-Test	AI-SGL	30	64.50	6.067	-1.228
	NAI-SGL	30	66.17	4.292	

Note. AI-SGL = artificial intelligence–serious game-based learning; NAI-SGL = non-artificial intelligence–serious game-based learning; N = sample size; t = t-value from the independent samples t-test. N = X; t = X.

Following the instructional activity, the post-test mean score for the experimental group (AI-SGL) was 78.50 (SD = 6.180), while the mean score for the control group (NAI-SGL) was 75.83 (SD = 5.427), as shown in Table 4. Each instructional group consisted of 30 students (N=30). An analysis of covariance was conducted to examine whether there was a statistically significant difference in post-test scores between the two groups, controlling for the influence of prior knowledge (pre-test scores).

Table 4. Statistics for the post-test

		N	Mean	Standard Deviation
Post-Test	AI-SGL	30	78.50	6.180
	NAI-SGL	30	75.83	5.427

Note. AI-SGL = artificial intelligence–serious game-based learning; NAI-SGL = non-artificial intelligence–serious game-based learning; N = sample size. N = X.

The results of the analysis of covariance, presented in Table 5, indicate a significant main effect of the instructional group on post-test performance, $F(1, 57) = 10.444$, $p = 0.002$ ($p < 0.05$), which suggests that students in the AI-SGL group achieved significantly higher learning gains compared to those in the NAI-SGL group. In this analysis, the degrees of freedom (df) for the instructional group was 1, reflecting the comparison between AI-SGL and NAI-SGL, while the residual error had df = 57. The corrected model had df = 2, accounting for both the instructional group and the covariate (pre-test scores). These results underscore the effectiveness of integrating AI within the serious game-based learning environment in fostering greater knowledge acquisition.

Table 5. Tests of between-subjects effects

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	516.957[a]	2	447.260	21.719	<.001
Intercept	366.379	1	366.379	17.791	<.001
Pre	787.853	1	787.853	38.258	<.001
Class	215.081	1	215.081	10.444	.002
Error	1173.814	57	20.593		
Total	359350.000	60			
Corrected Total	2068.333	59			

Note. *F* = F-ratio from the analysis of variance (ANOVA); *Sig.* = significance level (p-value); *a* = indicates the model includes the covariate (pre-test score). df = degrees of freedom.

Analysis of Engagement

The study also investigated differences in student engagement between the two instructional groups. An independent samples t-test was conducted to assess whether the difference in engagement scores between the two groups was statistically significant. As shown in Table 6, the engagement analysis included 30 students in each group (N=30), the mean engagement score for the experimental group (AI-SGL) was 127.13 (SD = 14.263), while the control group (NAI-SGL) had a mean engagement score of 113.80 (SD = 15.128). The t-test for equality of means reveals a statistically significant difference in engagement scores, $t(58) = 3.512$, $p = 0.001$. These results suggest that students in the experimental group exhibit significantly higher levels of engagement compared to those in the control group. This finding highlights the potential of incorporating AI into serious game-based learning environments to foster greater student involvement and active participation.

Table 6. T-test results of motivation

		N	Mean	Standard Deviation	t
Motivation	AI-SGL	30	127.13	14.263	3.512
	NAI-SGL	30	113.80	15.128	

Note. AI-SGL = artificial intelligence–serious game-based learning; NAI-SGL = non-artificial intelligence–serious game-based learning; N = sample size; t = t-value from the independent samples t-test.
***$p < 0.001$.

DISCUSSION AND CONCLUSION

Discussion About Research Questions

The present study developed and implemented the learning serious game Mystery Patterns to explore the effectiveness of integrating AI virtual characters in enhancing design history education through role-playing narratives. The findings reveal that students in the AI-SGL group demonstrated significantly higher levels of knowledge acquisition and engagement compared to their counterparts in the NAI-SGL group.

In response to research question Q1, the findings of this study indicate that integrating AI virtual characters in serious games significantly improves students' knowledge acquisition compared to traditional serious games. This effect can be attributed to personalized feedback, tailored learning paths, and contextual guidance facilitated by AI-driven interactions. The use of AI virtual characters in Mystery Patterns allowed for personalized feedback, tailored learning paths, and contextual guidance, which collectively contributed to a deeper understanding of design history concepts. These findings align with prior research highlighting the effectiveness of interactive and adaptive learning technologies. For example, Pescarin and Martinez Pandiani (2022) demonstrated that role-playing games set in immersive contexts can motivate students and improve knowledge retention in education. Similarly, Zhang et al. (2021) found that online role-playing games offer collaborative learning opportunities and enhance knowledge acquisition through interactive and narrative-driven environments. Furthermore, Naul and Liu (2019) emphasized the importance of empathetic characters and adaptive narratives in serious games for creating meaningful learning experiences. The personalized guidance offered by AI characters in the current study is consistent with these findings, as it enabled students to actively engage with the cultural and historical aspects of the arts and crafts movement.

Regarding Q2, the results demonstrate that AI virtual characters enhance student engagement by providing immersive, narrative-driven experiences. These findings highlight the potential of AI-driven role-playing games in increasing both motivation and interaction within educational settings. Students in the AI-SGL group reported higher levels of engagement, attributed to the immersive and narrative-driven learning experience provided by the virtual characters. These characters acted as facilitators, creating a sense of companionship and emotional connection, which motivated students to participate more actively in tasks such as artifact bidding, archival investigation, and pattern reproduction. This aligns with the findings of Zhan et al. (2022), who identified role-playing games as effective tools for fostering engagement through simulation and interaction. Moreover, Fernes et al. (2024) highlighted the value of immersive serious games in increasing user engagement and knowledge of historical topics by placing students in realistic, context-rich scenarios. The narrative design and adaptive feedback mechanisms in Mystery Patterns further resonate with the conclusion of Alrehaili and Osman (2019) that role-playing in serious games promotes higher motivation and engagement compared to traditional methods.

The findings of this study underscore the value of integrating virtual characters into serious games to enhance both knowledge acquisition and engagement. The AI-SGL system provided a more interactive and learner-centered experience, leveraging narrative and role-based mechanics to support

educational outcomes. These results align with broader trends in educational game design, emphasizing the significance of personalized and immersive learning environments.

Limitations and Future Study

However, several limitations of this study should be acknowledged. First, the implementation of AI virtual characters requires substantial computational resources and technical expertise, which may limit the accessibility of such systems in resource-constrained educational settings. Second, the current system focuses specifically on the arts and crafts movement, and its effectiveness in teaching other design history periods remains untested.

Third, while this study employed a controlled trial, its moderate sample size (N = 60) limits the generalizability of the findings. Additionally, the pre- and post-tests used identical items, with randomization serving as the primary measure to mitigate memory recall bias. Although this approach helps reduce test-related confounds, familiarity with the test items may still have influenced the results. Future studies should consider incorporating alternative test versions or additional measures to further control for this potential effect.

Fourth, the intervention was relatively short, lasting only 100 min in a single session. While significant learning gains were observed immediately after gameplay, long-term retention was not assessed. In contrast, many educational interventions—particularly in disciplines like language learning—span weeks or months before measurable learning gains can be fully evaluated. To enhance the robustness of the findings, future research should incorporate a delayed post-test (e.g., one month after the intervention) to assess knowledge retention over time.

Therefore, improvement of the proposed approach may be attributed to the effective integration of AI virtual characters within the gaming scenarios and historical exploration tasks. The virtual characters' ability to provide personalized guidance and contextual feedback made students more receptive to the design history content during the gaming process. This finding offers valuable insights for developers and educators seeking to create educational games with intelligent learning support tools for design history education. Future research should explore the adaptation of this AI-supported approach to different design history contexts and movements. Additionally, longitudinal studies should examine the long-term retention of historical knowledge acquired through such gaming experiences.

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